

iQOO HACKATHON 2026 • SMARTPHONE AI INNOVATION

ChainTrace AI

On-Device Multimodal Forensic Intelligence & Autonomous VASP Attribution

Turning fragmented crypto evidence into legally ready freeze packets in under 2 minutes.

GitHub: <https://github.com/abhinav2007-cyber/chaintrace.ai> | Working Prototype & AI Engine

Crypto Crimes Move at Blockchain Speed. Investigators Don't.

Law enforcement and anti-fraud investigators face insurmountable friction during the first 60 golden minutes.

01

The 'Golden Hour' Freeze Dilemma

Once victim funds are transferred, scammers bridge, hop, and cash out via centralized exchanges (VASPs) within 15-45 minutes. Traditional forensic workflows take 24-72 hours to trace, missing the freeze window entirely.

02

Fragmented Multimodal Evidence

Victims submit screenshots, phone chat logs, suspicious wallet addresses, transaction hashes, and bank slips. Officers have to manually copy-paste, correlate, and verify every piece across multiple tools.

03

Cloud Privacy & Jurisdictional Friction

Uploading sensitive evidentiary screenshots and victim PII to public cloud LLMs violates evidence chain-of-custody and data privacy laws. On-device local processing is a non-negotiable imperative.

THE INNOVATION

ChainTrace AI: Autonomous Investigation at the Edge

A full-stack, on-device intelligence platform designed to empower frontline officers.

OCR & ENTITY FUSION • Multimodal Ingestion

Upload chat screenshots, victim complaints, raw wallet strings, and TX hashes. The engine automatically extracts and cross-correlates cryptographic entities.

AUTONOMOUS GRAPHING • Live Investigation Graph

Constructs a real-time directed graph connecting victims, scammer wallets, intermediate hops, peel chains, and target VASP deposit nodes with visual risk coloring.

VASP ATTRIBUTION ENGINE • Explainable Attribution

Identifies destination exchange deposit clusters (Binance, Bybit, OKX, WazirX) with confidence percentages and human-readable evidence backing.

COURT-READY ARTIFACTS • 1-Click Freeze Packets

Instantly generates formal Section 91 CrPC notice letters and legal freeze requests complete with SHA-256 evidence integrity hashes and audit logs.

End-to-End On-Device Forensic Pipeline

Zero external cloud dependency: from local evidence capture to legal warrant packet.

STEP 1

Evidence Ingestion

Mobile camera capture, WhatsApp chats, transaction hashes, wallet addresses.

STEP 2

AI Parsing & OCR

Local entity extraction, regex address validation, amount & timestamp normalization.

STEP 3

Knowledge Graph

NetworkX & graph correlation, multi-hop trace, risk cluster classification.

STEP 4

VASP Attribution

Deposit address matching, exchange cluster confidence scoring, explainable reasoning.

STEP 5

Legal Freeze Output

Automated formal statutory notices, SHA-256 audit ledger, PDF export.

WORKING PROTOTYPE

Key Platform Features in the Working Model

Live at <http://localhost:3000> & fully functional API at <http://localhost:8000>

★ Interactive Multi-Hop Graph

Interactive visual graph rendering scammer nodes, intermediate hops, mixer interaction, and VASP exit wallets. Users can click nodes to inspect transaction volumes and timestamps.

★ Explainable 'Why?' Engine

Never a black box: every VASP attribution explains the mathematical heuristic and heuristics evidence (e.g. 94% confidence: 2 hops from scammer, 14.5 ETH deposit pattern).

★ ChainTrace Copilot (Chat)

Interactive AI assistant directly accessible in the UI. Investigators can query in plain English: 'Which exchange has the largest deposit?' or 'Draft an inquiry for Binance compliance'.

★ Instant Law Enforcement Packets

One-click generation of court-ready legal notices with case IDs, officer credentials, target wallet addresses, and formal freeze instructions.

Why the iQOO Flagship Architecture is Essential

Harnessing Qualcomm Snapdragon NPU, Monster Performance Engine & On-Device Security.

ADVANTAGE 01

Edge NPU AI Processing

iQOO's high-efficiency Neural Processing Unit runs local OCR and entity recognition models directly on-device in milliseconds without needing internet connectivity or cloud servers.

ADVANTAGE 02

Air-Gapped Evidence Privacy

Crime scene screenshots and victim personal identities never leave the phone. Meets strict legal standards for digital forensics and data sovereignty.

ADVANTAGE 03

Sub-2-Minute Field Triage

Frontline cyber officers can interview victims, scan their mobile wallets, and initiate an emergency VASP freeze notice right from their iQOO smartphone in the field.

ENGINEERING RIGOR

Production-Grade Modern Tech Stack

Built with high performance, modularity, and rapid scalability in mind.

Frontend Application

Next.js 14, React, Tailwind CSS, Lucide Icons, Obsidian Ember Theme

Backend Services

FastAPI (Python 3.11), Uvicorn, RESTful architecture, Pydantic v2 schemas

Graph & Heuristics

NetworkX topological trace engine, multi-hop flow calculation, clustering

On-Device & AI Integration

Local multimodal parsing, explainable attribution rationale engine, copilot agent

Storage & Cryptography

SQLite local DB, SHA-256 evidentiary chain-of-custody verification

Security & Compliance

Zero cloud data leakage, offline-first execution, automated legal notice generation

Why ChainTrace AI Stands Out

A real, deployable tool solving a billion-dollar law enforcement problem.

01

Working Prototype Ready

Not just slides or mockups — live frontend, backend, graph engine, and AI copilot.

02

Democratizing Forensics

Turns multi-thousand-dollar desktop enterprise software into an agile smartphone tool.

03

Ready for Android / iQOO APK

Engineered to compile directly as an on-device Android application powered by ONNX/TFLite.

04

Open Source on GitHub

Fully tracked and documented repository at github.com/abhinav2007-cyber/chaintrace.ai